

KUSHAL CHATURVEDI

AI Engineer | Intelligent Systems | Workflow Automation

Hyderabad, Telangana, India

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PROFILE SUMMARY

- AI Engineer with **4.5+ years of experience** designing and delivering **AI-powered workflow automation systems**, intelligent data platforms, and scalable orchestration architectures.
- Hands-on experience using **Large Language Models (Claude Sonnet)** for AI-assisted system design, code generation, refactoring, and technical documentation.
- Strong background in **Python, Prefect, metadata-driven orchestration**, and parallel execution frameworks enabling zero-code and low-code workflow deployment.
- Proven ability to modernize legacy **COBOL/JCL mainframe workloads** into Python-based intelligent pipelines with measurable performance and maintainability gains.

EXPERIENCE

AI Engineer – Intelligent Workflow & Automation Systems

Aug 2023 – Present

Cognizant Technology Solutions

Hyderabad, India

- Designed and developed an **AI-powered workflow orchestration platform** using **Python, Prefect 2.x, YAML-based metadata**, and **Claude Sonnet LLM**.
- Utilized **Claude Sonnet** for AI-assisted code generation, architectural reasoning, optimization, refactoring, and documentation, reducing development effort by **60%**.
- Engineered a **configuration-driven execution framework** enabling dynamic task creation, dependency resolution, and rule-based validations for business workflows.
- Implemented **parallel multi-workflow execution** supporting **10+ concurrent workflows** with intelligent retry mechanisms and failure categorization.
- Modernized complex **COBOL/JCL mainframe ETL pipelines** by translating business logic into Python, achieving **40% faster processing**.
- Built enterprise-grade data quality frameworks including schema enforcement, encoding detection (UTF-8/Latin-1), deduplication, and integrity checks processing **500K+ records/day**.

Data Engineer – Mainframe ETL

July 2021 – Aug 2023

Cognizant Technology Solutions

Hyderabad, India

- Designed and supported large-scale batch ETL pipelines on IBM z/OS using **COBOL, JCL, VSAM, DB2, and SYNCSORT**.
- Implemented delta processing, deduplication, reconciliations, and multi-file joins for regulatory and operational reporting systems.
- Contributed to modernization initiatives by collaborating with Python and cloud engineering teams to migrate core business logic.

FEATURED AI PROJECTS

AI-Powered Intelligent Workflow Orchestration Platform

AI Engineer

- Developed an enterprise-grade orchestration system integrating **Claude Sonnet LLM** with Prefect to design, generate, and validate workflows dynamically.
- Implemented YAML-driven dynamic task generation enabling zero-code workflow onboarding for non-technical users.
- Achieved **99.5% production uptime** using intelligent retries, execution analytics, and observability on Prefect Cloud.

High-Performance Data Reconciliation Engine

AI Engineer / Developer

- Built a scalable reconciliation engine using **Python, DuckDB, Pandas, and Parquet** to compare datasets with **5M+ rows**.
- Optimized SQL-driven execution achieving **15x performance improvement**, reducing reconciliation time from hours to minutes.
- Generated automated Excel reports with conditional formatting for enterprise data validation and audit workflows.

TECHNICAL SKILLS

AI & LLMs: Claude Sonnet, Prompt Engineering, AI-Assisted Development, LLM-Based Design

Workflow Orchestration: Prefect 2.x, Dynamic Workflow Generation, Parallel Execution

Programming: Python (asyncio, multiprocessing, Pandas, DuckDB, SQLAlchemy, Streamlit)

Data Engineering: PySpark, Spark SQL, ETL/ELT, Parquet, CSV, JSON, Fixed-width

Databases: PostgreSQL, DuckDB, MySQL, DB2, MongoDB, VSAM

Document Intelligence: PyMuPDF (fitz), PDF Text Extraction, Visual Diffing

Cloud & Tools: Azure, Databricks, ADF, ADLS Gen2, Git, CI/CD, VS Code

EDUCATION

Jamia Millia Islamia

Sep 2017 – May 2021

B.Tech in Mechanical Engineering (9.6 SGPA)

New Delhi, India